Report on *"Matrix Theory Reloaded: A BPS Road to Holography"*

(Blair, Lahnsteiner, Obers, Yan)

Summary of the paper. The paper treats the decoupling limit that gives matrix theory on Dp-branes as a *BPS decoupling limit* of the whole of type II string theory. This limit defines "matrix p-brane theory" (MpT), whose ten-dimensional target space is non-Lorentzian, with a codimension-$(p+1)$ foliation and p-brane Galilean boosts. The authors then make three claims:

(i) applying the MpT limit at the asymptotic infinity of the Dp-brane supergravity solution induces exactly the Maldacena/IMSY near-horizon limit. Applying two such limits, aligned with different branes, produces non-Lorentzian bulk geometries, among them the proposals of Lambert *et al.*;

(ii) after uplifting to M-theory, each BPS limit is a DLCQ. This leads to the conjecture that holography in general is a DLCQ^n/DLCQ^m correspondence ($m > n$): AdS$_5$/CFT$_4$ is DLCQ^0/DLCQ^1 and D1–D5 is DLCQ^0/DLCQ^2;

(iii) running the limit in reverse is a p-brane generalisation of the $T\bar{T}$ deformation. This gives explicit classical flow equations: free scalars flow to Nambu–Goto in any dimension, the Maxwell-plus-scalars equations hold for $p = 1, 2$, and only perturbative results are given for $p \geq 3$. When the parameter is made radius dependent, $\omega \to \Omega(r)$, the deformation gives the near-horizon bulk metric.

1. Principal achievement

The main achievement is to unify the subject: a single operation organises results that were previously separate. The BPS (critical RR-field) decoupling limit defines matrix theory on flat D-branes. Applied to the curved brane solution, the *same* limit is the near-horizon limit. The paper shows that the limit at fixed α', with the critical $C^{(p+1)}$, is equivalent to the Maldacena scaling ($g_{\rm YM}$ and $v = R/\alpha'$ fixed). It also explains why the near-horizon geometry approaches the flat MpT geometry again as $r/\ell \to \infty$: the conformal factor $\Omega(r)$ plays the role of ω there. This answers a clear question, namely which geometry $\mathcal{N} = 4$ SYM "sees" at infinity.

More importantly, the framework is *generative*. Requiring the limit to stay finite gives simple selection rules: $q = p + 4$ for aligned branes, and $p + q = 10$ with $m + n = 4$ in general. These rules recover the web of non-Lorentzian dualities of Lambert *et al.*, which had been constructed bottom-up, and place them alongside AdS/CFT, multicritical limits, non-relativistic M-theory and spin matrix theory in one duality web. The most concrete new result is the flow equation

$$\partial_\lambda \mathcal{L} = \tfrac{1}{2\lambda^2}\Big\{\mathrm{tr}(\mathbb{1} - \lambda\mathcal{T}) - (p-1)\big[\det(\mathbb{1} - \lambda\mathcal{T})\big]^{1/(p-1)} - 2\Big\}.$$

It deforms D free scalars in $p+1$ dimensions into the static-gauge Nambu–Goto p-brane. It reduces to $T\bar{T}$ for $p = 1$ and to the Gross *et al.* quantum-mechanical deformation for $p = 0$, and it gives a simple membrane equation for $p = 2$. It also explains why ordinary $T\bar{T}$ applied to 2D Maxwell theory does not give DBI. All of this can be checked independently.

2. Principal drawback

The main weakness is a gap between kinematics and dynamics. Almost all of the evidence for the central holographic claims consists of matching scalings of classical supergravity solutions and worldvolume actions. No observable is computed and compared on both sides of any new duality. Specifically:

- For $n = 0$ the authors themselves describe the result as a rephrasing of the standard story. The new content is the case $n \geq 1$, and there the non-Lorentzian bulk backgrounds are *not shown to solve* the MpT supergravity equations or their torsion constraints, which are not yet formulated. Several of these backgrounds also need an *ad hoc* rescaling $\ell^{7-q} \propto \omega^{\#}$ (effectively a large-N scaling) whose physical meaning the authors say is unclear.
- The DLCQ^n/DLCQ^m conjecture is not stated sharply. No dictionary is given: there is no statement of which bulk fields source which operators, nor of what is supposed to be equal. The inductive support is thin. $n = 0$ is known; $n = 1$ rests on proposals that are themselves largely untested; $n = 2$ relies on spin matrix theory, whose third DLCQ "may be associated"; for $n \geq 3$ there are no examples. As it stands the conjecture works as a classification scheme more than as a falsifiable statement.
- The "intrinsic" generation of the bulk by $T\bar{T}$ takes $\Omega(r)$ as input from the known solution. Promoting $\omega \to \Omega(r)$ uses a dilatation symmetry, but nothing in the deformation fixes Ω. The flow equations are classical, bosonic and abelian, and the F-dependent equations are complete only for $p \leq 2$. None of the quantum properties that make $T\bar{T}$ special (solvable spectrum, factorisation) is established for $p > 1$.

3. Recommended follow-up: a dynamical test of DLCQ^1/DLCQ^2 via the M-theory uplift

Idea. Take the paper's flagship non-Lorentzian example: the M0T limit of the D4-brane near-horizon geometry, conjectured to be dual to the M0T limit of 5D SYM (equivalently the null reduction of the 6D $(2, 0)$ theory, i.e. quantum mechanics on instanton moduli space). Then *exploit the uplift.* M0T is the IIA reduction of DLCQ M-theory, and the D4 lifts to an M5 wrapping the null circle. So the bulk is the near-horizon M5 geometry $\mathrm{AdS}_7 \times S^4$ with a lightlike circle along the boundary. This turns the open question of whether the bulk solves M0T supergravity into a statement about a known Lorentzian solution. It also removes the strong coupling at infinity for $p = 4$ that the authors themselves flag. With this in hand one can compute on both sides:

(a) *Symmetries.* Compute the Killing (super)algebra of the non-Lorentzian background (the $p = 0$, $q = 4$ case of the paper's near-horizon solution) in terms of (τ, E, φ, c). Show that it equals the centraliser of the null momentum in $\mathfrak{osp}(8^*|4)$, a Schrödinger-type superalgebra, and match it to the symmetries of the field theory. The authors list this computation as "natural and important".

(b) *Dictionary.* Show that the flat M0T data (τ, E) at asymptotic infinity act as sources for the field theory's non-relativistic stress tensor ("non-Lorentzian GKPW"). This would make precise the paper's claim that the SYM couples to the asymptotic non-Lorentzian geometry.

(c) *Protected spectrum and thermodynamics.* Compare the supergravity Kaluza–Klein spectrum at fixed null momentum N with BPS states of the instanton quantum mechanics. Construct the non-extremal (black-brane) version of the limit, which also touches the paper's suggested finite-temperature direction, and check the N^3-type entropy scaling.

Why this one. *Feasibility:* every ingredient exists already. The $\mathrm{AdS}_7 \times S^4$ Kaluza–Klein spectrum, the superalgebra and the near-extremal M5 solution are standard; the limits are algebraic; and

step (a) alone is a self-contained result. *Impact:* it would be the first test of the conjecture that goes beyond limits. If it succeeds, the non-Lorentzian geometry becomes a working holographic tool and the inductive step from $n = 0$ to $n = 1$ gains real support. If it fails, for example through the known subtleties of null compactifications (zero modes, a small lightlike circle), we learn where the DLCQ^n picture breaks. Either result is informative. The recipe then carries over by duality to the other members of the web, such as the M2-brane and non-relativistic-string examples. The alternatives score lower on this balance. Quantising the new $p = 2$ membrane $T\bar{T}$ flow could have high impact but is technically uncertain, and writing down the general MpT supergravity equations is a large and somewhat open-ended task. The uplift route gets a dynamical test without needing either.